EM-mixture component

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$K=2 \quad \theta = (\pi_1, \mu_1, \mu_2, \sigma_1^2, \sigma_2^2)$

$$Q(\theta,\theta^0) = E_{Z|X,\theta^0}\left[\log p(X_1,\dots X_n, Z_1 \dots Z_n|\theta)\right]$$

$$= \sum_{i=1}^{n}\Big[p(Z_i=1|X,\theta^0)\Big(-\log\sqrt{2\pi\sigma_1^2} - \frac{1}{2}\frac{(x_i-\mu_1)^2}{\sigma_1^2} + \log\pi_1\Big)$$

$$+ p(Z_i=2|X,\theta^0)\Big(-\log\sqrt{2\pi\sigma_2^2} - \frac{1}{2}\frac{(x_i-\mu_2)^2}{\sigma_2^2} + \log(1-\pi_1)\Big)\Big]$$

$$\frac{\partial Q(\theta,\theta^0)}{\partial\pi_1} = \sum_{i=1}^{n}\left[\frac{p(Z_i=1|X,\theta^0)}{\pi_1} - \frac{p(Z_i=2|X,\theta^0)}{1-\pi_1}\right] = 0$$

$$\frac{(1-\pi_1)\left[\sum_{i=1}^{n} p(Z_i=1|X,\theta^0)\right] - \pi_1\left[\sum_{i=1}^{n} p(Z_i=2|X,\theta^0)\right]}{\pi_1(1-\pi_1)} = 0$$

$$\hat{\pi}_1 = \frac{\sum_{i=1}^{n} p(Z_i=1|X,\theta^0)}{\sum_{i=1}^{n}\Big[\underbrace{p(Z_i=1|X,\theta^0) + p(Z_i=2|X,\theta^0)}_{=1}\Big]}$$

$$= \frac{\sum_{i=1}^{n} p(Z_i=1|X,\theta^0)}{n}$$

$K=3. \quad \theta = (\pi_1, \pi_2, \mu_1, \mu_2, \mu_3, \sigma_1^2, \sigma_2^2, \sigma_3^2)$

$$Q(\theta,\theta^0) = E_{Z|X,\theta^0}\left[\log p(X_1,\dots X_n, Z_1 \dots Z_n|\theta)\right]$$

$$= \sum_{i=1}^{n}\Big[p(Z_i=1|X,\theta^0)\Big(-\log\sqrt{2\pi\sigma_1^2} - \frac{1}{2}\frac{(x_i-\mu_1)^2}{\sigma_1^2} + \log\pi_1\Big)$$

$$+ p(Z_i=2|X,\theta^0)\Big(-\log\sqrt{2\pi\sigma_2^2} - \frac{1}{2}\frac{(x_i-\mu_2)^2}{\sigma_2^2} + \log\pi_2\Big)$$

$$+ p(Z_i=3|X,\theta^0)\Big(-\log\sqrt{2\pi\sigma_3^2} - \frac{1}{2}\frac{(x_i-\mu_3)^2}{\sigma_3^2} + \log(1-\pi_1-\pi_2)\Big)\Big]$$

$$\frac{\partial Q(\theta,\theta^0)}{\partial\pi_1} = \sum_{i=1}^{n}\left[\frac{p(Z_i=1|X,\theta^0)}{\pi_1} - \frac{p(Z_i=3|X,\theta^0)}{1-\pi_1-\pi_2}\right]$$

$$\frac{\boxed{(1-\pi_1-\pi_2)\left[\sum_{i=1}^{n} p(Z_i=1|X,\theta^0)\right] - \pi_1\left[\sum_{i=1}^{n} p(Z_i=3|X,\theta^0)\right]}}{\pi_1(1-\pi_1-\pi_2)} = 0 \quad (1)$$

$$\frac{\partial Q(\theta,\theta^0)}{\partial\pi_2} = \sum_{i=1}^{n}\left[\frac{p(Z_i=2|X,\theta^0)}{\pi_2} - \frac{p(Z_i=3|X,\theta^0)}{1-\pi_1-\pi_2}\right]$$

$$\frac{\boxed{(1-\pi_1-\pi_2)\left[\sum_{i=1}^{n} p(Z_i=2|X,\theta^0)\right] - \pi_2\left[\sum_{i=1}^{n} p(Z_i=3|X,\theta^0)\right]}}{\pi_2(1-\pi_1-\pi_2)} = 0 \quad (2)$$

from (1) $\Rightarrow$ $\boxed{(1-\pi_1-\pi_2)\left[\sum_{i=1}^{n} p(Z_i=1|X,\theta^0)\right] = \pi_1\left[\sum_{i=1}^{n} p(Z_i=3|X,\theta^0)\right]}$ $\leftarrow (1)'$

from (2) $\Rightarrow$ $\boxed{(1-\pi_1-\pi_2)\left[\sum_{i=1}^{n} p(Z_i=2|X,\theta^0)\right] = \pi_2\left[\sum_{i=1}^{n} p(Z_i=3|X,\theta^0)\right]}$ $\leftarrow (2)'$

$(1)' + (2)' \Rightarrow (1-\pi_1-\pi_2)\left[\sum_{i=1}^{n}\big(p(Z_i=1|X,\theta^0) + p(Z_i=2|X,\theta^0)\big)\right]$

$$= \underbrace{(\pi_1+\pi_2)}_{1-(1-\pi_1-\pi_2)}\left[\sum_{i=1}^{n} p(Z_i=3|X,\theta^0)\right]$$

$$1-\pi_1-\pi_2 = \frac{\sum_{i=1}^{n} p(Z_i=3|X,\theta^0)}{\sum_{i=1}^{n}\Big(\underbrace{p(Z_i=1|X,\theta^0) + p(Z_i=2|X,\theta^0) + p(Z_i=3|X,\theta^0)}_{=1}\Big)}$$

$$= \frac{\sum_{i=1}^{n} p(Z_i=3|X,\theta^0)}{n}$$

from $(1)'$ $\Rightarrow$

$$\hat{\pi}_1 = (1-\pi_1-\pi_2)\frac{\left[\sum_{i=1}^{n} p(Z_i=1|X,\theta^0)\right]}{\left[\sum_{i=1}^{n} p(Z_i=3|X,\theta^0)\right]}$$

$$= \frac{\cancel{\sum_{i=1}^{n} p(Z_i=3|X,\theta^0)}}{n}\frac{\left[\sum_{i=1}^{n} p(Z_i=1|X,\theta^0)\right]}{\cancel{\left[\sum_{i=1}^{n} p(Z_i=3|X,\theta^0)\right]}}$$

$$= \frac{\sum_{i=1}^{n} p(Z_i=1|X,\theta^0)}{n}$$

from $(2)'$ $\Rightarrow$

$$\hat{\pi}_2 = (1-\pi_1-\pi_2)\frac{\left[\sum_{i=1}^{n} p(Z_i=2|X,\theta^0)\right]}{\left[\sum_{i=1}^{n} p(Z_i=3|X,\theta^0)\right]}$$

$$= \frac{\cancel{\sum_{i=1}^{n} p(Z_i=3|X,\theta^0)}}{n}\frac{\left[\sum_{i=1}^{n} p(Z_i=2|X,\theta^0)\right]}{\cancel{\left[\sum_{i=1}^{n} p(Z_i=3|X,\theta^0)\right]}}$$

$$= \frac{\sum_{i=1}^{n} p(Z_i=2|X,\theta^0)}{n}$$